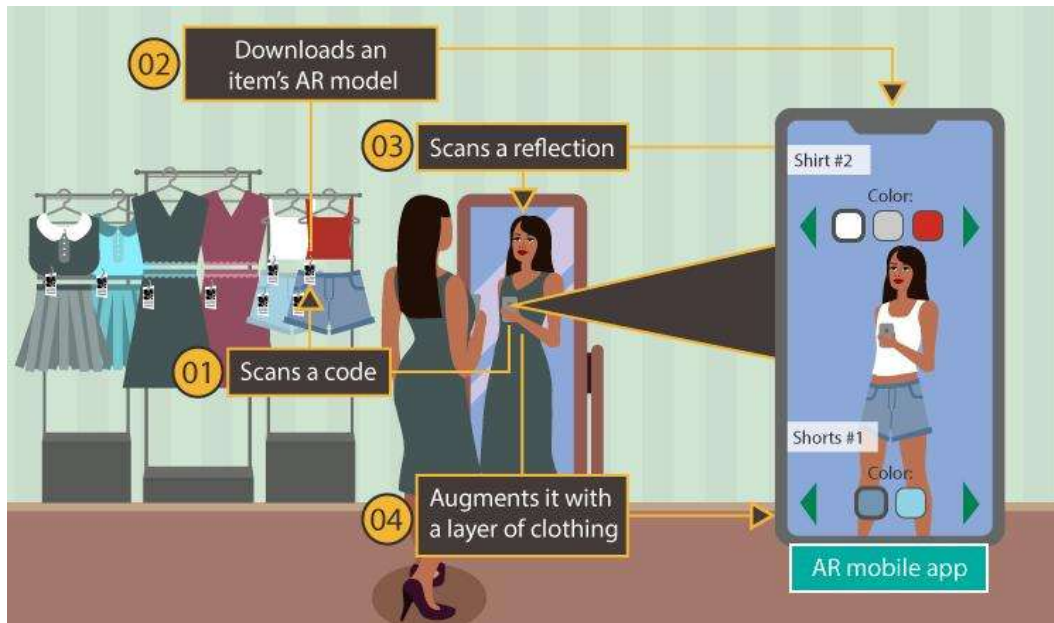


TASK 3 — AI Research

1. Virtual Try-On AI Systems



Overview

Virtual try-on systems allow users to visualize clothing on their body digitally using AI and computer vision. This eliminates the need to physically try clothes before purchasing.

How It Works

1. Detect human pose using AI
2. Identify body landmarks
3. Map clothing onto body
4. Adjust size and alignment
5. Render realistic output

Technologies Used

- Computer Vision
- Deep Learning
- Augmented Reality (AR)
- Pose Detection (e.g., MediaPipe)

Advantages

- Enhances user experience
- Saves time
- Reduces return rates
- Improves confidence in online shopping

2. Fashion Recommendation Models

Overview

Fashion recommendation systems use AI to suggest clothing items based on user preferences, trends, and behavior.

How It Works

- Collect user data:
 - Browsing history
 - Preferences
 - Purchase patterns
- Apply machine learning algorithms
- Recommend suitable items

Types of Recommendation Systems

1. Content-Based Filtering

- Suggests items similar to user's past choices

2. Collaborative Filtering

- Suggests items based on other users with similar tastes

Applications

- E-commerce platforms
- Fashion apps
- Personalized styling systems

Benefits of AI in Fashion

- Personalized shopping experience
- Faster product discovery
- Improved customer satisfaction
- Better inventory management

Conclusion

AI technologies such as virtual try-on systems and recommendation models are transforming the fashion industry. They provide personalized, efficient, and interactive shopping experiences, making online fashion platforms more intelligent and user-friendly.